

7000 Series Phase Noise Tester / Signal Source Analyzer



Features

- All-In-One Compact Measurement System
- Measurements Down to -190 dBc/Hz
- Offset Range from 0.01 Hz to 100 MHz
- Highest Flexibility & Dynamic Range By Selectable Internal or External References
- Programmable Low Noise Power Supplies
- Powerful GUI and Programming Interface
- External Battery Pack Option

Applications

- High-Speed Production Testing of Phase Noise
- VCO Testing
- Additive Phase Noise Characterization of Amplifiers, Transmitters, Mixers
- Ultra-Low Phase Noise Crystal Oscillator Analysis
- Time Stability Analysis of Clocks
- Aerospace and Defense



7000 Series Datasheet v.1.25

Phase Noise Tester / Signal Source Analyzer from
1 MHz to 7, 26, or 40 GHz

DEFINITIONS

- The specifications in the following pages describe the warranted performance of the instrument for $23 \pm 5^\circ\text{C}$ after a 30-minute warm-up period (unless otherwise stated).

Min/Max: Parameter range that is guaranteed by product design, and/or production tested. Warranted performance specifications include guard-bands to account for the expected statistical performance distribution, measurement uncertainties, and changes in performance due to environmental conditions.

Typical: Expected mean values, not warranted performance.

INTRODUCTION

• Fully integrated cross-correlation signal source analyzer for 1 MHz to 7, 26, and 40 GHz

The 7000 Series is an integrated solution that offers an indispensable set of measurement functions for evaluating signal sources ranging from VHF to microwave frequencies such as crystal oscillators, PLL synthesizers, clocks, phase-locked or free-running VCOs, DROs, SAW or YIG oscillators, and others.

The flexible instrument comprises a two-channel cross-correlation system with two internal tuneable reference sources and also allows measurements with externally fed references.

The 7000 Series supports many other functions including

- Absolute and residual phase noise measurements
- Amplitude noise measurements
- Pulsed absolute and residual phase noise measurements
- Two-channel 100 MHz FFT analysis
- Transient measurements (frequency, phase, amplitude vs. time)
- Spectrum analysis
- Frequency counter function / power meter

Additionally, the unit offers

- Two programmable low noise DC supplies up to 15 V and 550 mA current capability
- Three low noise tuning voltages for -5 to +20 V voltage range

It is a compact and powerful instrument available with LAN (VXI-11), USBTMC, or with GPIB (optionally) interfaces. Platform independent intuitive graphical user interface (GUI), API library, and powerful SCPI command language set is available.

Operated with an external 24 V DC supply, it consumes less than 70 W.

SPECIFICATIONS

Absolute Phase Noise Measurement 1 MHz to 40 GHz (continuous waveform)

Measurement parameters:

SSB Phase Noise [dBc/Hz]

Spurious Noise [dBc]

Integrated RMS Phase Noise Deviation [deg, rad]

Time Jitter [s]

Residual FM/PM [Hz RMS]

PARAMETER	MIN	TYPICAL	MAX	NOTE
RF Frequency Range	<i>FMIN</i> 1 MHz 1 MHz 1 MHz		<i>FMAX</i> 7 GHz 26 GHz 40 GHz	Using internal references 7070 7300 7340
	5 MHz 5 MHz		7 GHz 18 GHz	Using external references 7070 7300 / 7340
Input Power Range				Damage level +26 dBm
<18 GHz	-15 dBm		+20 dBm	
18 GHz to 30 GHz	-15 dBm		+23 dBm	See RF sensitivity plots
> 30 GHz	-5 dBm		+23 dBm	
Input Impedance		50 Ω		AC coupled, 10 V DC max
VSWR		2		
Offset Analysis Range	0.01 Hz 0.01 Hz		100 MHz > 25% of f_c	$f_c > 150$ MHz $f_c < 150$ MHz
Resolution (PPD)	200	200	1600	RBW adjustable (x1/x2/x4/x8), PPD (points per decade) can be lower for lowest decade of measurement
Measurement Accuracy		± 4 dB ± 3 dB ± 2 dB		Offset < 10 Hz Offset 10 Hz to 1 kHz Offset 1 kHz to 100 MHz
Phase Noise Sensitivity				See plot & sensitivity tables
Spurious Levels				
Internal References		-90 dBc		
External References		-85 dBc		
Measurement Time				See table "Measurement Time"
Trigger				Single, continuous, manual, bus
Internal References				Cross-correlation
Frequency Range	1 MHz		<i>FMAX</i>	
Phase Noise Sensitivity				See plots "Sensitivity"
RF Tracking Range		± 1 ppm ± 10 ppm ± 1000 ppm		Option LN Standard High drift mode
External References				single channel / cross-corr.

Frequency Range	5 MHz 5 MHz		7 GHz 18 GHz	7070 7300 / 7340
RF Input Power Range	0 dBm		+23 dBm	Damage level +26 dBm
Phase Noise Sensitivity				See plot & sensitivity tables
Reference Input Level Range				
< 1.3 GHz	+10 dBm	+15 dBm	+21 dBm	Lower input ports
> 1.3 GHz	+13 dBm	+15 dBm	+21 dBm	Upper input ports
Tuning Voltage Range	-5 V		+20 V	User adjustable
Tuning Output Current			20 mA	

Absolute Phase Noise Measurement – Pulsed (Option PULSE / NPS)

PARAMETER	MIN	TYPICAL	MAX	NOTE
RF Frequency Range	30 MHz 30 MHz		7 GHz 26 GHz 40 GHz	7070 7300 7340
RF Input Power Range	+5 dBm		+20 dBm	No power measurement
Input Parameters				
Pulse Rate (PRF)	300 Hz		2 MHz	Option PULSE Option NPS Option PULSE Option NPS
Pulse Width	1 us 50 ns		2 ms 2 ms	
Duty Cycle	2% 0.1%		60% 60%	
Offset Analysis Range	0.01 Hz		PRF	
Measurement Accuracy		±4 dB ±3 dB ±2 dB		Offset < 10 Hz Offset 10 Hz to 1 kHz Offset 1 kHz to 100 MHz
Measurement Time				See table “Meas. Time”

Residual (Additive) Phase Noise Measurement – CW (Option APN) and Pulsed (Option APN + PULSE)

Measurement parameters:

SSB Phase Noise [dBc/Hz]

Spurious Noise [dBc]

Integrated RMS Phase Noise Deviation [deg, rad]

Time Jitter [s]

Residual FM/PM [Hz RMS]

PARAMETER	MIN	TYPICAL	MAX	NOTE
RF Frequency Range	10 MHz 10 MHz		7 GHz 18 GHz	7070 7300 / 7340
RF Input Power Range				
RF Port	+3 dBm		+23 dBm	Option LO
REF Ports	+13 dBm		+20 dBm	
LO Output Power Range	+17 dBm		+23 dBm	
Offset Analysis Range	0.01 Hz		100 MHz	
Measurement Accuracy		±3 dB ±2 dB		Offset < 1 kHz Offset > 1 kHz

Additive Phase Noise Sensitivity				See sensitivity table
----------------------------------	--	--	--	-----------------------



Transient Analysis (Option TRAN)

Measurement parameters:

Wideband Mode (WB): Frequency [Hz]

Narrowband Mode (NB): Frequency [Hz], RF Power [dB], Phase [deg]

PARAMETER	MIN	TYPICAL	MAX	NOTE
RF Frequency Bands (WB)	5 MHz 20 MHz 80 MHz 320 MHz 1.3 GHz 5.2 GHz		100 MHz 400 MHz 1.6 GHz 3 GHz 26 GHz FMAX	Band 1 Band 2 Band 3 Band 4 Band 5 Band 6
Measurement Spans				
Wideband Mode (WB)				Bands 1-6
Narrowband Mode (NB)	200 kHz		80 MHz	200 kHz, 1.25 MHz, 80 MHz
Frequency Resolution				See table
Time Span	10 μ s		1 min	
Time Resolution	16 ns		50 ms	
Trigger Mode				Single, Continuous, Internal (WB video or NB video), external



Burst Mode Phase Noise Measurement (Option PULSE + Option BURST)

Measurement parameters:

SSB Phase Noise [dBc/Hz]

PARAMETER	MIN	TYPICAL	MAX	NOTE
RF Frequency Range	5 MHz		FMAX	
Offset Analysis Range	1 / T		30 MHz	
Time Span (T)	10 μ s		1 min	
Phase Noise Sensitivity				Single Channel, f = 1 GHz
1 kHz		-120 dBc/Hz		
10 kHz		-128 dBc/Hz		
100 kHz		-131 dBc/Hz		
1 MHz		-131 dBc/Hz		
10 MHz		-147 dBc/Hz		



Absolute Amplitude Noise Measurement (Option AM)

Measurement parameters:

SSB Amplitude Noise [dBc/Hz]

PARAMETER	MIN	TYPICAL	MAX	NOTE
RF Frequency Range	5 MHz		7 GHz 26 GHz 40 GHz	7070 7300 7340
RF Input Power Range				
5 MHz to 10 GHz	-20 dBm		+20 dBm	
10 GHz to 40 GHz	-10 dBm		+20 dBm	

Offset Analysis Range	0.1 Hz		40 MHz	
Measurement Uncertainty		±2 dB		
AM Noise Sensitivity (1 corr.)				1 GHz, P _{in} = -10 dBm to +20 dBm
1 Hz		-100 dBc/Hz		
10 Hz		-115 dBc/Hz		
100 Hz		-135 dBc/Hz		
1 kHz		-145 dBc/Hz		
10 kHz		-155 dBc/Hz		
> 100 kHz		-160 dBc/Hz		



Baseband Noise Analysis

Input Connectors:

2 BNC female (rear panel), AC coupled

Measurement parameters:

Noise Spectrum [nV/√Hz]

PARAMETER	MIN	TYPICAL	MAX	NOTE
Frequency Input Range	1 Hz		100 MHz	
DC Voltage Input Range	-12 V		+12 V	
Input Impedance		1 kΩ		DC
AC Voltage Range			+10 dBm	
Input Noise Density (1 correlation)				
10 kHz		< 3 nV/√Hz		
Trigger				Single, Continuous, Manual, Bus



Time Stability Measurement (Option TSTAB)

Measurement parameters:

ADEV (no dead time)

PARAMETER	MIN	TYPICAL	MAX	NOTE
Measurement Time	1 s		10 days	
Resolution Bandwidth (RBW)	1 Hz	100 Hz	100 Hz	setable to 1 Hz, 100 Hz
ADEV Sensitivity				With RBW 100 Hz
τ = 1 s		5e-13		
τ = 100 s		1e-13		

Spectrum Monitoring (Option SPEC)

Measurement parameters:

Spectral Noise Density [dBm]

PARAMETER	MIN	TYPICAL	MAX	NOTE
RF Frequency Range	5 MHz		7 GHz	7070
	5 MHz		26 GHz	7300
	5 MHz		40 GHz	7340
RBW	5.8 Hz		58 kHz	
Measurement Uncertainty				
Absolute		±3 dB		
Relative		±1 dB		
Noise Floor				
10 MHz to 4 GHz		-95 dBm/Hz		
4 GHz to 18 GHz		-90 dBm/Hz		
18 GHz to 40 GHz		-80 dBm/Hz		
Spurious (SFDR)				Spurious Free Dynamic Range
10 MHz to 4 GHz		-70 dBc		
4 GHz to 18 GHz		-60 dBc		
18 GHz to 40 GHz		-55 dBc		
Spurious (absolute)		dBm		
10 MHz to 4 GHz				
4 GHz to 18 GHz				
18 GHz to 40 GHz				
Trigger				Continuous

VCO Characterization (Option VCO)

Measurement parameters:

Frequency [Hz]

K_{VCO} (Tuning Sensitivity) ($\Delta f / \Delta V_c$) [Hz/V]

Frequency Pushing [Hz/V]

RF Power Level [dBm]

DC Supply Current [mA]

SSB Phase Noise [dBc/Hz]

PARAMETER	MIN	TYPICAL	MAX	NOTE
Sweep Parameters				
DC Supply Voltage	0 V		15 V	Adjustable
DC Supply Current			550 mA	
Tuning Voltage	-5 V		20 V	Adjustable
Tuning Current			20 mA	
RF Frequency Range	5 MHz		F_{MAX}	
Uncertainty		0.5 ppm		
RF Input Power Range	-5 dBm		+20 dBm	
Uncertainty		0.5 dB	2 dB	
DC Supply Current	0 mA		550 mA	
Uncertainty		1%		
Output Settling Time		20 ms		
Measurement Speed		70 ms / point		Frequency, K_{VCO} , Pushing, Supply Current, and Power

Frequency Counter

Measurement parameters:

Frequency [Hz]

PARAMETER	MIN	TYPICAL	MAX	NOTE
RF Frequency Range	1 MHz		<i>F</i> MAX	
Absolute Accuracy		300 ppb		Or accuracy of external reference
Sensitivity				See plot "Typical RF Sensitivity Plot"

Power Detector

Measurement parameters:

RF Power Level [mW, dBm]

PARAMETER	MIN	TYPICAL	MAX	NOTE
RF Frequency Range	5 MHz 5 MHz		<i>F</i> MAX 40 GHz	7070, 7300 7340
Absolute Accuracy		±1 dB	±2.5 dB	
Power Range	-10 dBm		+13 dBm	

Tuning Voltage & Dual Power Supply

PARAMETER	MIN	TYPICAL	MAX	NOTE
DUT Tuning				BNC Front Panel Output
DC Voltage Range	-5 V		+22 V	
Setting Resolution		1 mV		
Setting Uncertainty		±2 mV		
Noise Level		< 2 nV _{rms} /√Hz		> 2 kHz
DC Current Range	0 mA		20 mA	
DC Power Supplies				BNC Rear Panel Output (Channel 1 & 2)
DC Voltage Range	0 V		15 V	
Setting Resolution		10 mV		
Setting Accuracy		±10 mV		
Noise Level		< 10 nV _{rms} /√Hz		> 20 kHz
Output Resistance		< 0.5 Ω		
DC Current Measurement Range	0 mA		550 mA	Per Channel
Resolution		100 μA		



LO Output (Option LO)

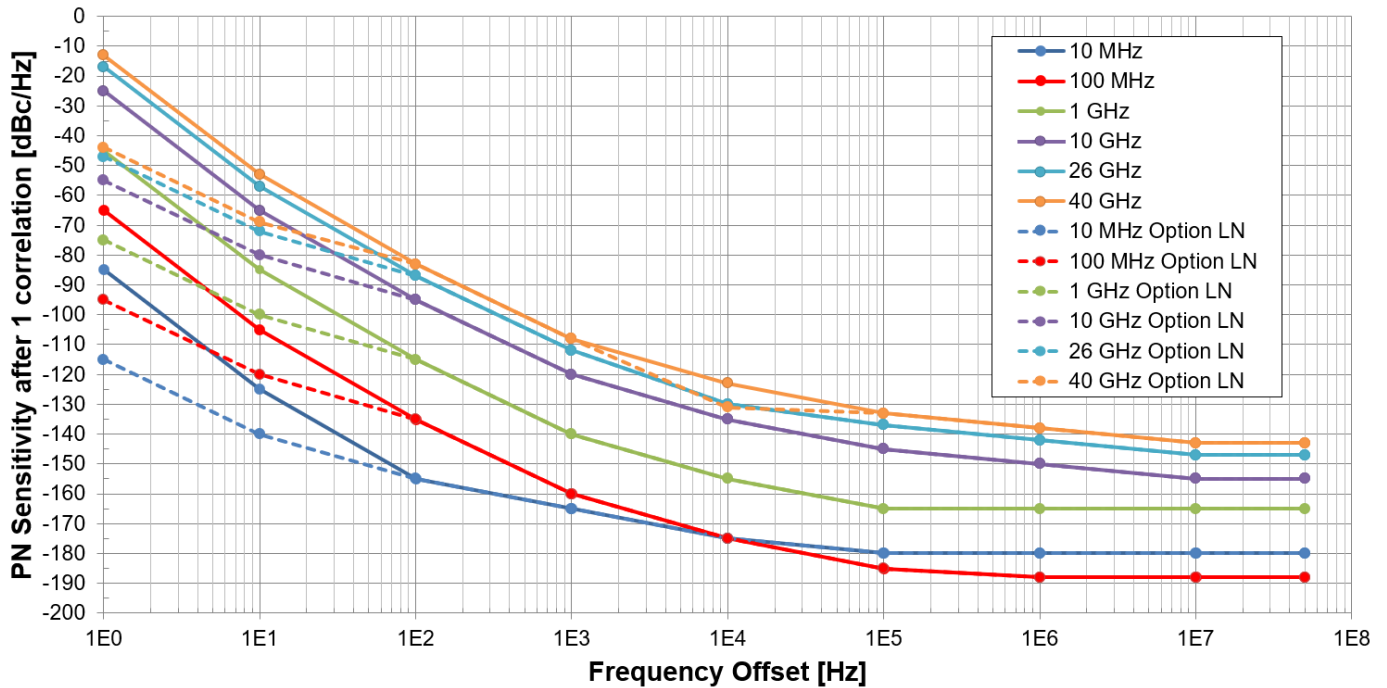
PARAMETER	MIN	TYPICAL	MAX	NOTE
Use: Additive Phase Noise				
Frequency Range	0 GHz		18 GHz	
Frequency Resolution		1 Hz		
Power Level	14 dBm	17.5 dBm	20 dBm	
Use: LO for Downconversion				
Frequency Range	2 GHz		20 GHz	
Frequency Resolution		0.5 GHz		
Power Level	14 dBm	17.5 dBm	21 dBm	

PERFORMANCE CURVES

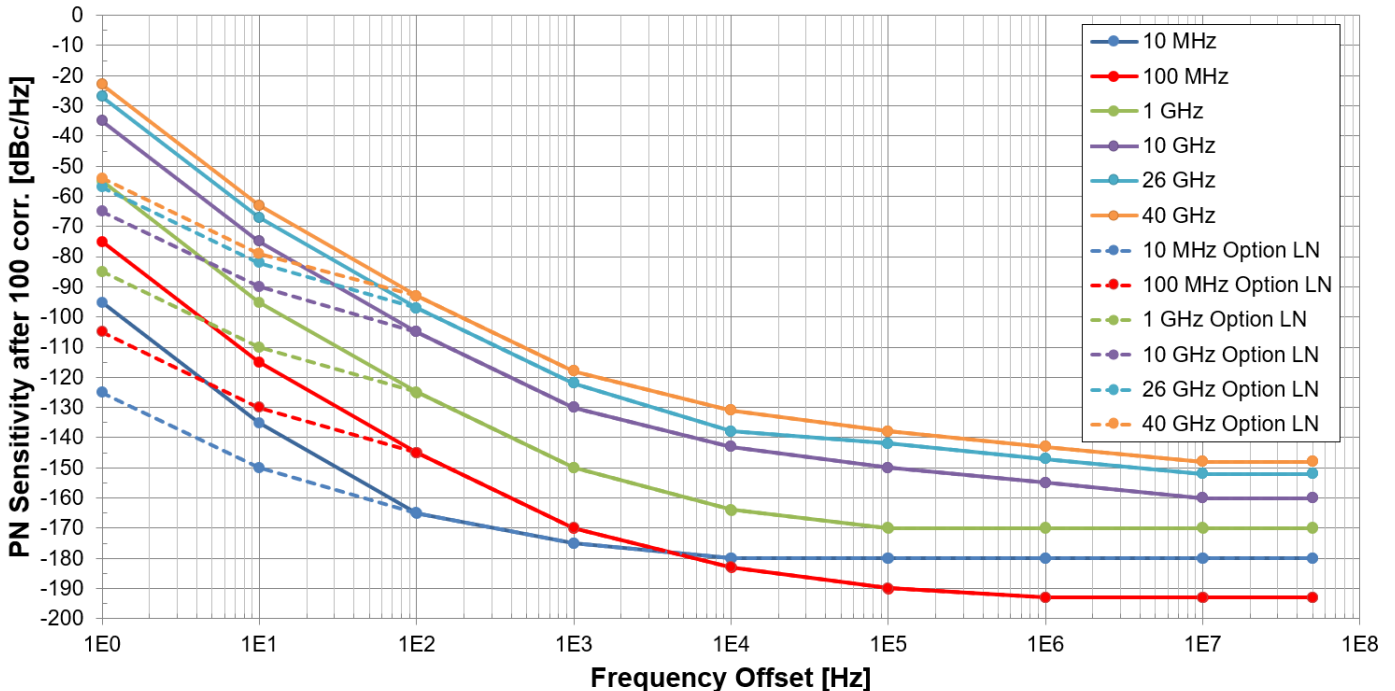
Phase Noise Sensitivity - Standard and Low Noise (Option LN) Internal References

Measurement Time ~10 seconds, after first cross-correlation; further correlations will improve sensitivity by 5 dB for 10, 10 dB for 100, and 15 dB for 1000 correlations performed. The plots show typical performance. Thermal noise (-177 dBm/Hz PM) can limit the measurement as well for low input power levels (<10 dBm).

After 1 Correlation

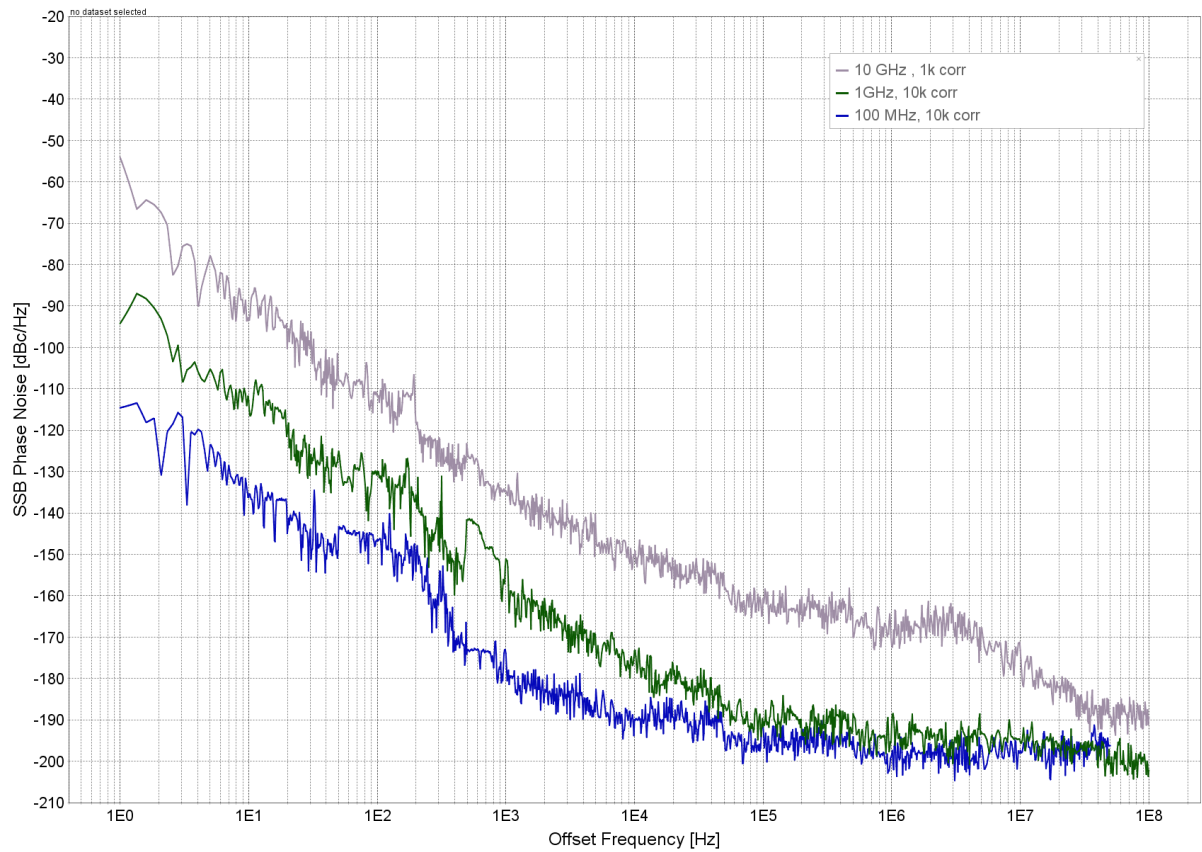


After 100 Correlations

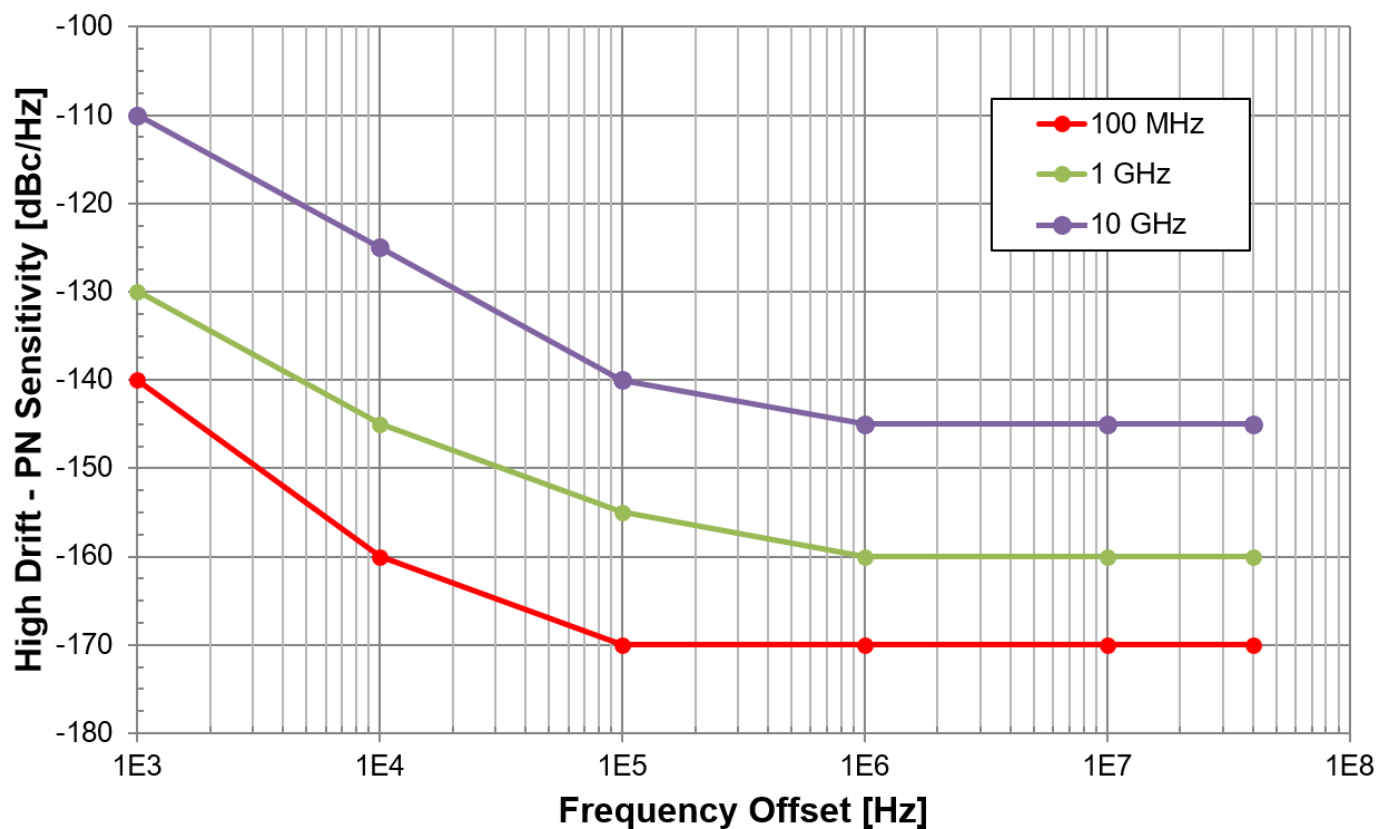




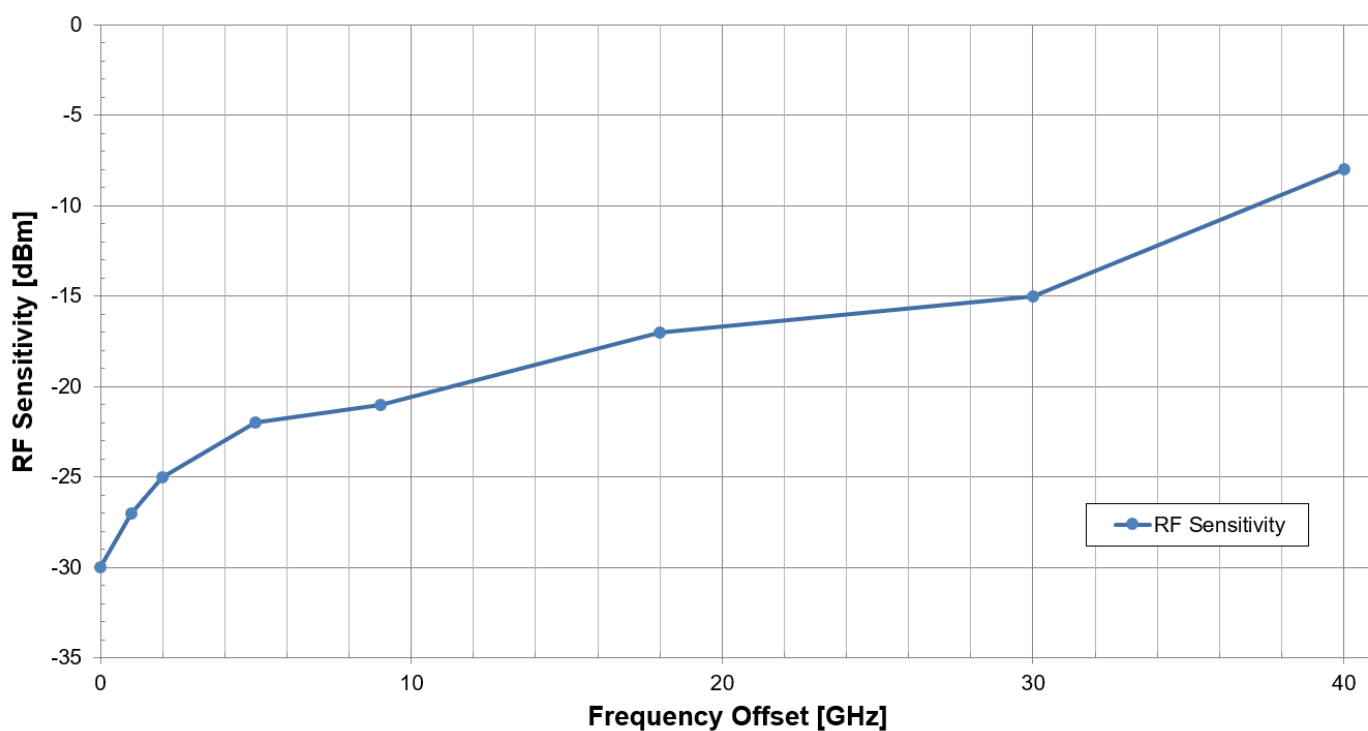
Typical Noisefloor Example (after >1k correlations at 100MHz, 1GHz, 10GHz)



Phase Noise Sensitivity - High Drift



Typical RF Sensitivity 5 MHz to 40 GHz (blue trace, in dBm)





Phase Noise Measurement Time

Total measurement time consists of setup time, transfer time plus the number of performed correlations times the time per correlation. The measurement times below are normalized to one correlation for nominal RBW settings per correlation and measurement times > 2 seconds.

	TIME PER CORRELATIONS	DEFAULT NR. OF POINTS (SETTABLE)
0.1 Hz to 100 MHz	80	250 per decade
1 Hz to 100 MHz	8	250 per decade
10 Hz to 100 MHz	0.8	250 per decade
100 Hz to 100 MHz	0.1	250 per decade
1 kHz to 100 MHz	0.01	250 per decade
10 kHz to 100 MHz	< 0.004	250 per decade



Absolute Phase Noise Sensitivity – Internal References (Standard)

Abs. PN with internal references (Option LN)	OFFSET						
	1 Hz	10 Hz	100 Hz	1 kHz	10 kHz	100 kHz	1 MHz
10 MHz	-85	-125	-155	-165	-172	-175	-175
100 MHz	-65	-105	-135	-160	-172	-178	-178
1 GHz	-45	-85	-115	-140	-155	-160	-160
3 GHz	-35	-75	-105	-130	-145	-150	-155
10 GHz	-25	-65	-95	-120	-135	-140	-145
25 GHz	-15	-55	-85	-110	-130	-135	-140
40 GHz	-13	-53	-83	-108	-123	-133	-138
Remarks	Test conditions: carrier power $\geq$ 5 dBm; after one correlation						



Absolute Phase Noise Sensitivity – Internal References (with Option LN)

Abs. PN with internal references (Option LN)	OFFSET						
	1 Hz	10 Hz	100 Hz	1 kHz	10 kHz	100 kHz	1 MHz
10 MHz	-115	-140	-155	-165	-172	-175	-175
100 MHz	-95	-120	-135	-160	-172	-178	-178
1 GHz	-75	-100	-115	-140	-155	-160	-160
3 GHz	-65	-90	-105	-130	-145	-150	-155
10 GHz	-55	-80	-95	-120	-135	-140	-145
25 GHz	-45	-70	-85	-110	-130	-135	-140
40 GHz	-44	-68	-83	-108	-123	-133	-138
Remarks	Test conditions: carrier power $\geq$ 5 dBm; after one correlation						

Absolute Phase Noise Sensitivity – External References

Abs. PN with external references	OFFSET						
	1 Hz	10 Hz	100 Hz	1 kHz	10 kHz	100 kHz	1 MHz
10 MHz	-135	-150	-155	-170	-175	-175	-175
100 MHz	-120	-130	-140	-170	-178	-178	-178
1 GHz	-100	-110	-125	-155	-170	-170	-170
3 GHz	-95	-110	-125	-155	-170	-170	-170
10 GHz	-90	-110	-120	-145	-155	-155	-155
18 GHz	-85	-105	-115	-130	-140	-145	-145
Remarks	Test conditions: carrier power ≥ 5 dBm; after one correlation						

Additive Phase Noise Sensitivity – Single Channel

Additive PN (1 channel)	OFFSET						
	1 Hz	10 Hz	100 Hz	1 kHz	10 kHz	100 kHz	1 MHz
$10 \text{ MHz} \leq f \leq 1 \text{ GHz}$	-130	-140	-150	-160	-170	-170	-170
$1 \text{ GHz} \leq f \leq 4 \text{ GHz}$	-130	-140	-150	-160	-170	-170	-170
$4 \text{ GHz} \leq f \leq 16 \text{ GHz}$	-115	-125	-135	-145	-150	-155	-160
Remarks	Test conditions: RF carrier power ≥ 10 dBm; REF ≥ 13 dBm Two channel cross-correlation can improve noise floor by 5 dB per 10x correlations.						



Transient Analysis – Wideband: Frequency Resolution vs. Time Resolution (residual FM, 5% video bandwidth, typical)

Frequency Measurement uncertainty is $\pm$ (resolution + time-base uncertainty). Tabulated resolutions are measured with the 7000 Series and DUT locked to the same 10 MHz reference. Input level 0 dBm.

Time Resolution	16 ns	128 ns	500 ns	1 μ s	$\geq 10 \mu$ s
Frequency Band	Frequency Resolution [Hz]				
5 MHz to 100 MHz	3 k	100	30	15	10
20 MHz to 400 MHz	5 k	700	200	100	20
80 MHz to 1.6 GHz	10 k	1 k	200	100	50
320 MHz to 3 GHz	30 k	1.5 k	300	150	150
1.3 GHz to 26 GHz	100 k	6 k	2 k	1 k	1 k
5.2 GHz to <i>FMAX</i>	500 k	20 k	4 k	2 k	2 k



Transient Analysis – Narrowband: Frequency Resolution vs. Time Resolution (residual FM, 80 MHz span, 5% video bandwidth, typical)

Frequency Measurement uncertainty is $\pm$ (resolution + time-base uncertainty). Tabulated resolutions are measured with the 7000 Series and DUT locked to the same 10 MHz reference. Input level 0 dBm.

Time Resolution	16 ns	128 ns	500 ns	1 μ s	10 μ s	$\geq 20 \mu$ s
Frequency Range	Frequency Resolution [Hz]					
< 200 MHz	1.5 k	50	10	4	4	4
< 800 MHz	2.5 k	150	15	10	4	4
< 2 GHz	2.5 k	500	20	10	4	4
< 20 GHz	30 k	4 k	150	70	20	7
> 20 GHz	50 k	4 k	400	150	50	15



Transient Analysis – Narrowband: Frequency Resolution vs. Time Resolution (residual FM, 1.25 MHz span, no video bandwidth, typical)

Frequency Measurement uncertainty is $\pm$ (resolution + time-base uncertainty). Tabulated resolutions are measured with the 7000 Series and DUT locked to the same 10 MHz reference. Input level 0 dBm.

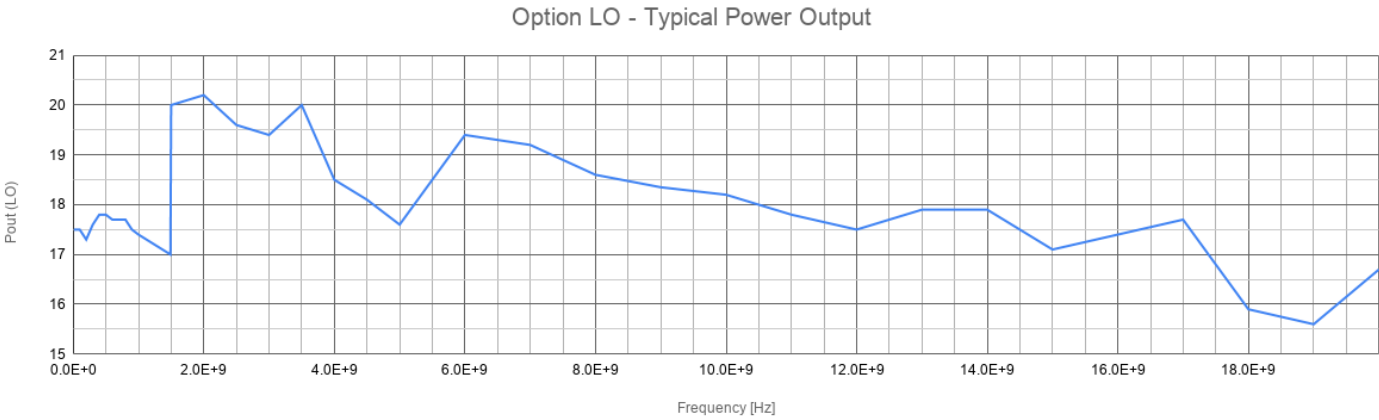
Time Resolution	256 ns	500 ns	1 μ s	10 μ s	$\geq 20 \mu$ s
Frequency Range	Frequency Resolution [Hz]				
< 200 MHz	60	30	15	1.5	0.5
< 800 MHz	70	30	15	1.5	1.5
< 2 GHz	100	40	15	3	1.5
< 20 GHz	1 k	300	150	30	15
> 20 GHz	3 k	1 k	400	60	30

Transient Analysis – Narrowband: Frequency Resolution vs. Time Resolution (residual FM, 200 kHz span, no video bandwidth, typical)

Frequency Measurement uncertainty is $\pm$ (resolution + time-base uncertainty). Tabulated resolutions are measured with the 7000 Series and DUT locked to the same 10 MHz reference. Input level 0 dBm.

Time Resolution	1 μ s	10 μ s	$\geq$ 20 μ s
Frequency Range	Frequency Resolution [Hz]		
< 200 MHz	1	0.5	0.3
< 800 MHz	1.5	0.5	0.3
< 2 GHz	3	1	0.4
< 20 GHz	20	10	3
> 20 GHz	50	20	10

Option LO – Power Output



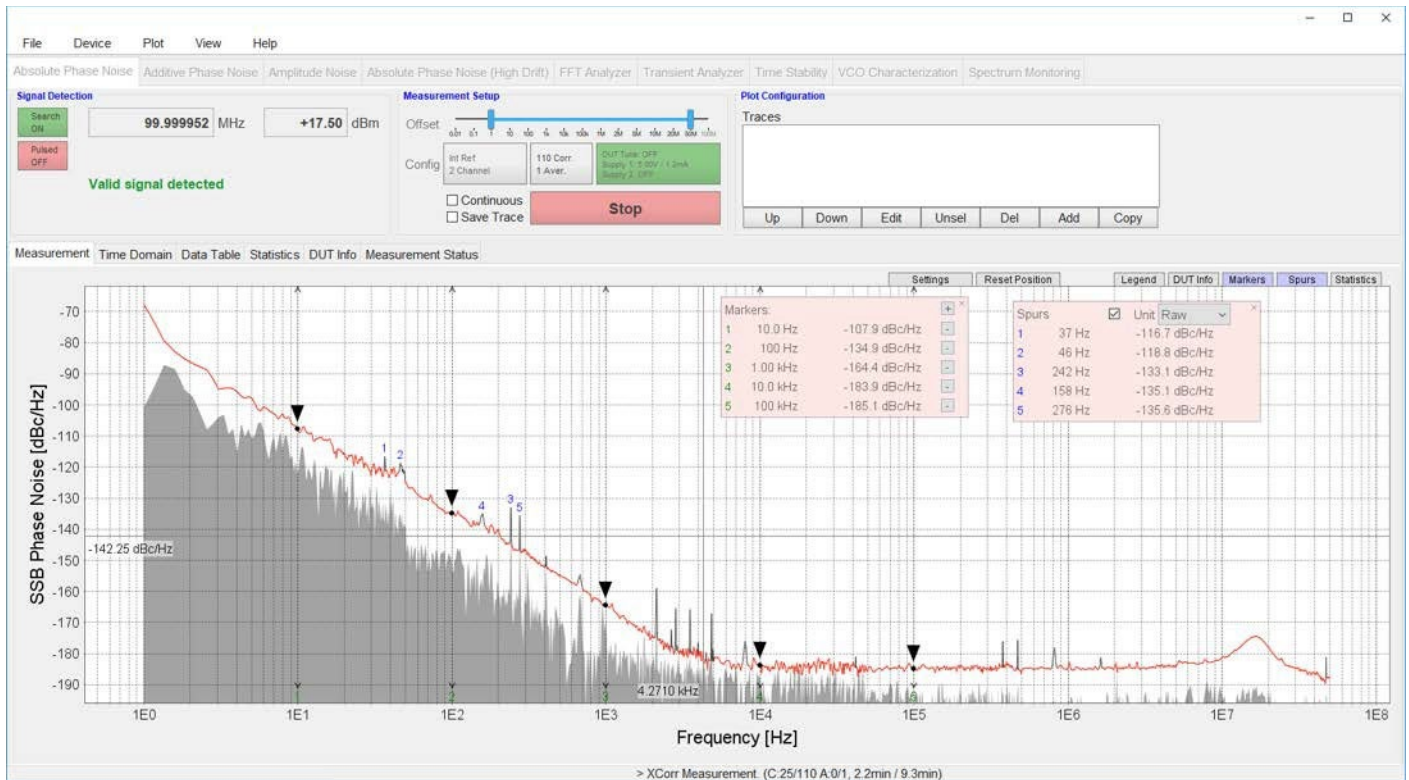


Data Processing Capabilities

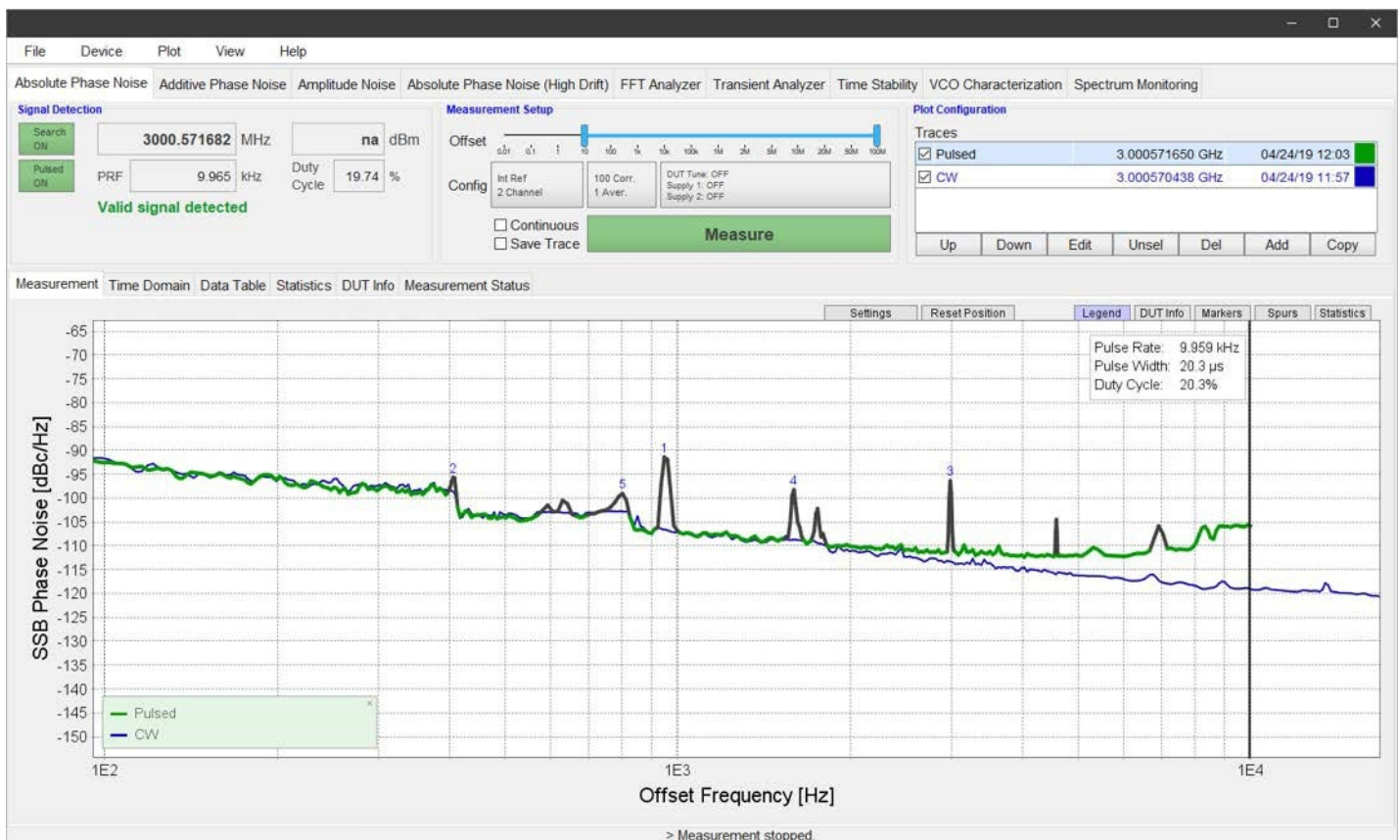
Graphical User Interface: The analyzer employs a graphical user interface based on the Windows operating system.

Display Functions	Phase Noise, Time Domain, Data Table, Residual, Statistics
Trace Functions	
Data Traces	Display current measurement and/or multiple memory data (up to 16 traces)
Title	Add customized title to each measurement window
Auto-Scale	Automatically selects scale resolution and reference value to vertically center the trace
Statistics	Calculates and displays mean, standard deviation, and peak-to-peak deviation of the trace
Marker Functions	16 independent markers

GUI Interface (Absolute Phase Noise)

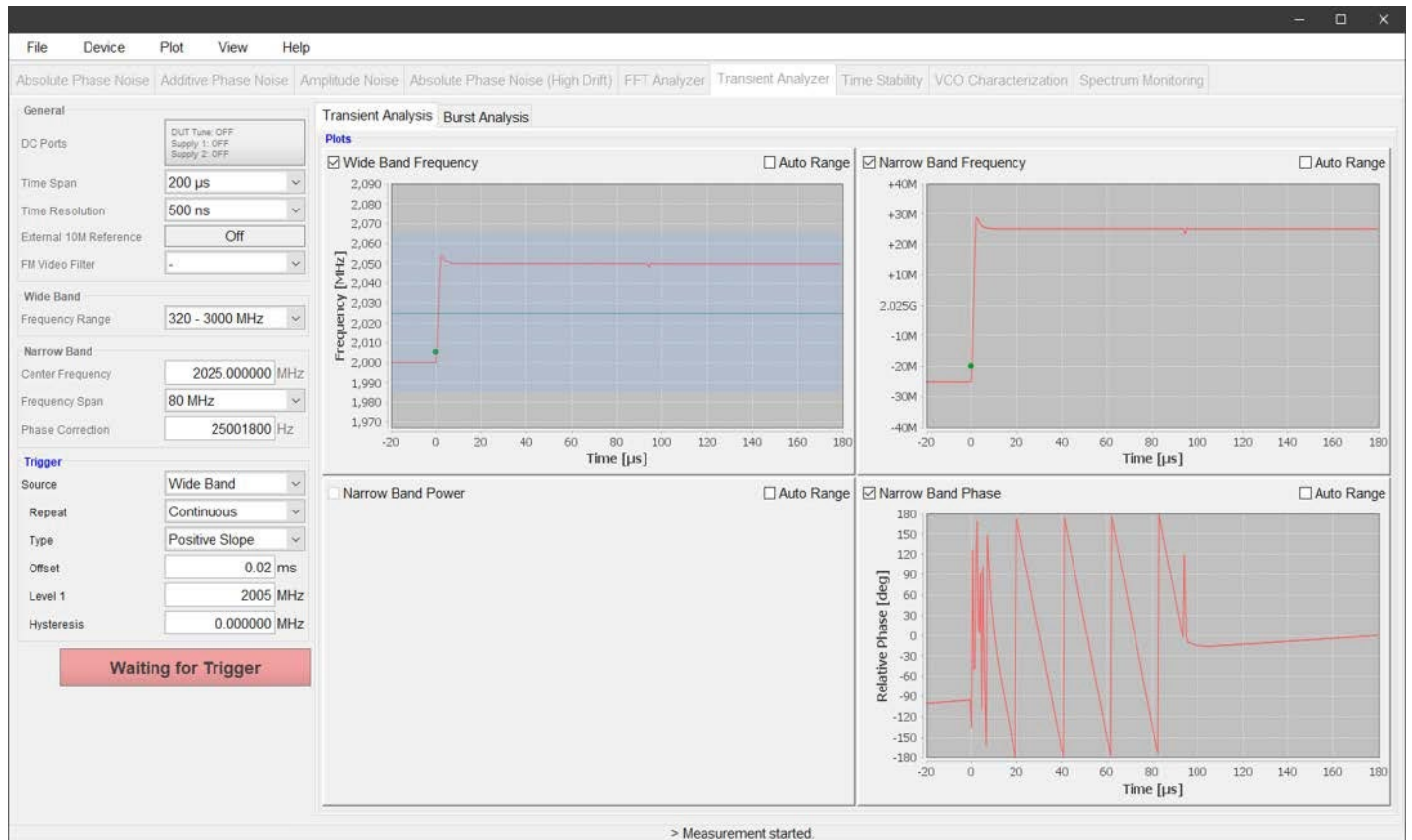


GUI Interface (PULSED RF Absolute Phase Noise)

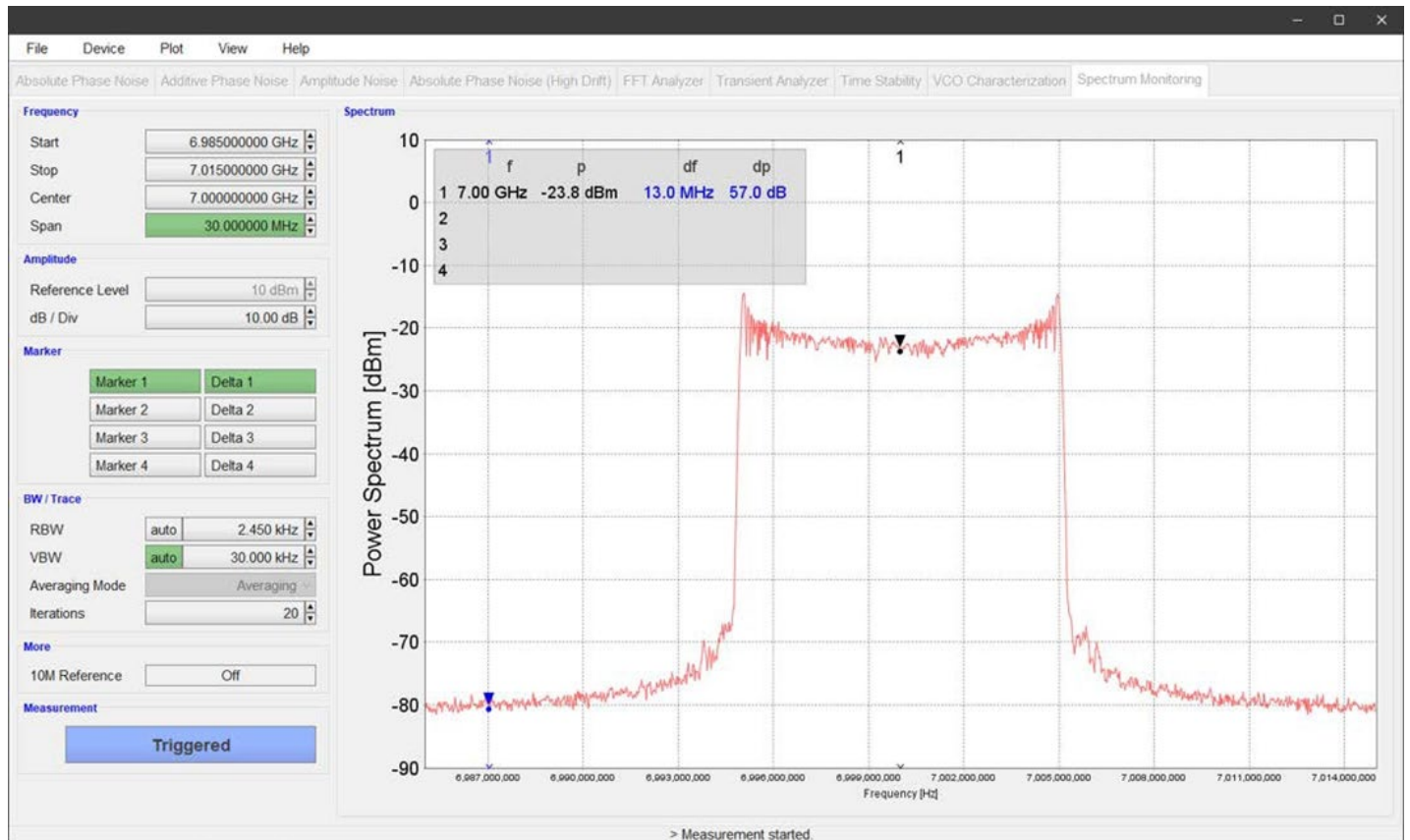


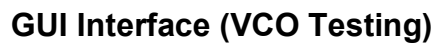


GUI Interface (Transient Analyzer)



GUI Interface (Spectrum Monitoring)





Connectors (Front)



RF Inputs

RF IN: SMA female (for 7070 / 7300); K female (for 7340)

REF1 IN HIGH/LOW, REF2 IN HIGH/LOW: SMA female

DC Outputs

REF1 TUNE, REF2 TUNE: BNC female

Operation

Switch I/O: DC Power Switch

POWER, READY, REMOTE: Status LED

Connectors (Rear)



HF/VHF/AUX Inputs

BASEBAND CH1, BASEBAND CH2: BNC female

REF IN 10 MHz: BNC female

EXT TRIG: BNC female

DC Outputs

DC SUPPLY CH1, DC SUPPLY CH2: BNC female

Operation

LAN: RJ-45

USB B: USB 2.0 device

DC 24V: DC Power Plug (24V, 2A)

GPIB (Option GPIB): IEEE-488 GPIB Connector

Connectors (Front – Option LO)



Additional RF Inputs

LO1 IN HIGH/LOW, LO2 IN HIGH/LOW: SMA female

RF1 IN, RF2 IN: SMA female

Additional RF Outputs

LO1 OUT HIGH/LOW, LO2 OUT HIGH/LOW: SMA female

RF1 OUT, RF2 OUT: SMA female

ORDERING INFORMATION



HOST MODEL	PRODUCT	DESCRIPTION
7000 Series	7070	Base Model: 7 GHz Signal Source Analyzer
7000 Series	7300	Base Model: 26 GHz Signal Source Analyzer
7000 Series	7340	Base Model: 40 GHz Signal Source Analyzer
7000 Series	Option LN	Ultra-low noise internal sources
7000 Series	Option PULSE	Pulsed signal measurement
7000 Series	Option NPS	Pulsed signal measurement for narrow pulses and low duty cycles
7000 Series	Option TRAN	Transient analysis
7000 Series	Option BURST	Burst mode phase noise measurement
7000 Series	Option AM	Amplitude noise measurement
7000 Series	Option APN	Additive phase noise measurement
7000 Series	Option LO	Access to internal reference for residual phase noise measurement (requires option APN)
7000 Series	Option TSTAB	Time stability analysis
7000 Series	Option VCO	VCO characterization
7000 Series	Option SPEC	Spectrum monitoring
7000 Series	Option GPIB	GPIB interface
	Accessory APNS	Traceable AM / PN noise standard, flange-mount module
	Accessory PS06	Mechanical phase shifter 1 – 6 GHz for additive phase noise measurements
	Accessory PS18	Mechanical phase shifter 6 – 18 GHz for additive phase noise measurements

GENERAL CHARACTERISTICS

Remote programming interfaces:

- Ethernet 100BaseT LAN interface
- USB 2.0 device
- GPB (IEEE-488.2,1987) with listen and talk (Option GPB)
- Control Language SCPI Version 1999.0

Power requirements: 24V $\pm$ 3.0 VDC; 70 W maximum

Mains adapter supplied: 100-240 VAC in / 24 V 4.0 A DC out

Environmental: Levels similar to MIL-PRF-28800F Class 3/4



notice

Safety / EMC complies with applicable Safety and EMC regulations and directives.

Weight: $\leq$ 10.0 kg (21 lbs) net

Dimensions:

- incl. rubber: 154 mm H x 467.5 mm W x 342 mm L (6.1 in H x 18.4 in W x 13.5 in L)
- with handle: 154 mm H x 520 mm W x 342 mm L (6.1 in H x 20.5 in W x 13.5 in L)
- handle: radius 230mm (9 in); can be turned 360° in 30° steps